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Foreword

The **Multidisciplinary International Theses and Articles Journal – Graduate Studies (2026)** presents a comprehensive collection of theses and scholarly articles that address contemporary issues in education, leadership, science, technology, and social development. This issue reflects the journal's commitment to advancing multidisciplinary research and providing a platform for graduate scholars, educators, researchers, and practitioners to disseminate evidence-based knowledge that informs academic, institutional, and professional practice.

The research studies featured in this volume examine diverse educational contexts, leadership practices, and instructional innovations. Among the highlighted works are *Decision-Making Competencies Correlate to Administrative Management for Teaching-Learning Enhancement in Zamboanga del Sur Division: An Expression of Extraordinary School Governance* by **Herlo D. Merto (2022)**, which explores the relationship between school leaders' decision-making competencies and effective administrative management, and an earlier study by the same author, *Classroom Social Environment and Student Participation in Technical-Vocational-Livelihood Education* (**Merto, 2019**), which investigates the influence of classroom climate on student engagement and participation.

Additional studies focus on financial literacy, instructional innovation, and technical education. These include *Financial Literacy and Financial Management Practices among Public Elementary School Teachers in Selected Schools in Nueva Vizcaya* by **Regino L. Galate et al.**, and *Utilization of an Electrical Wiring Board to Improve the Academic Performance of Grade 8 Students in Electrical Installation and Maintenance* by **Eduardo B. Somera**. Collectively, these investigations highlight the importance of practical interventions, teacher competencies, and innovative instructional resources in enhancing educational outcomes.

Complementing these graduate research studies are scholarly articles that explore sustainability, technology integration, leadership, values formation, and stakeholder collaboration in education. Featured contributions include *Green Chemistry: Advancing Science Through Sustainability* and *Effects of Google on Learning and Information Behavior* by **Prof. Cristopher G. Reyes**, which examine environmental responsibility and the evolving digital learning landscape. Other articles emphasize the significance of teacher–parent partnerships by **Catherine T. Galao**, leadership through exemplary practice by **Jomar E. Calibuso**, and values formation in basic education by **Ronelda D. Pahitong**.

This issue also features thought-provoking articles by **Rachelle F. Buncad**, addressing pressing educational and societal concerns through discussions on the Sustainable Development Goals (SDGs), gender equality in the primary grades, accountability in educational leadership, and ethical governance. These contributions reinforce the vital role of educators and school leaders in promoting inclusive, equitable, and values-driven educational environments.

Furthermore, the journal proudly includes the scholarly contributions of **Dr. Darlene Cabras-Derije**, namely *Transformational Leaders in the New Educational Scheme* and *School Principal's Impact Towards Learners' Scholastic Achievement*, which provide valuable insights into transformational leadership and its influence on school effectiveness and student academic success.

Overall, the **2026 Issue of the Multidisciplinary International Theses and Articles Journal – Graduate Studies** underscores the significance of multidisciplinary inquiry in addressing complex educational, organizational, technological, and societal challenges. The research and scholarly articles presented in this volume offer meaningful implications for educational policy, instructional practice, leadership development, and future research. It is our hope that this publication will serve as a valuable

resource for educators, researchers, policymakers, graduate students, and other stakeholders committed to advancing knowledge, fostering innovation, and promoting excellence in education and research.

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Part 1

Research Articles

DECISION-MAKING COMPETENCIES CORRELATE TO ADMINISTRATIVE MANAGEMENT FOR TEACHING- LEARNING ENHANCEMENT IN ZAMBOANGA DEL SUR DIVISION: AN EXPRESSION OF EXTRAORDINARY SCHOOL GOVERNANCE

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Modified Title: Decision-Making Competencies and Administrative
Management in Teaching– Learning Enhancement: Evidence of
Extraordinary School Governance in the Zamboanga del Sur Division

Abstract

This study examined the relationship between school administrators' decision-making competencies and administrative management in enhancing the teaching–learning process in the Zamboanga del Sur Division. Specifically, it assessed administrators' competencies in determining the type of decision, identifying necessary information,

involving stakeholders, establishing priorities, and anticipating decision outcomes, as well as their administrative management practices in planning, organizing, implementing, and coordinating school programs. A descriptive survey research design was employed using a self-developed, validated questionnaire administered through purposive sampling to school administrators and teachers with at least three years of service. Descriptive statistics, t-test, z-test, and correlation analysis were utilized to analyze the data. Findings revealed that school administrators consistently demonstrated high levels of decision-making competencies and administrative management practices, particularly in stakeholder involvement, organizing, implementation, and planning. Both administrators and teachers shared similar perceptions regarding these competencies and management practices, as evidenced by the absence of significant differences between their responses. Likewise, correlation analysis indicated no significant relationship between decision-making competencies and administrative management. Despite this, the consistent application of sound decision-making and effective management practices contributed positively to teaching–learning enhancement and reflected extraordinary school governance. The study concludes that inclusive decision-making, strategic planning, effective organization, and collaborative leadership remain essential in strengthening school governance and improving educational outcomes.

Keywords: decision-making competencies, administrative management, school leadership, teaching–learning enhancement, school governance, instructional leadership, participative decision-making, educational administration

Introduction

School administrators are expected to be exemplary leaders. They are entrusted with leadership roles because of their educational training, professional competence, experience, and ability to guide institutions toward excellence. As visionary leaders, school administrators are responsible for ensuring the continuous improvement of school programs and practices. They must possess a deep understanding of their school's past and present performance and be capable of clearly communicating their ideas and expectations to teachers and staff. Furthermore, administrators must remain responsive to the demands of a rapidly changing global environment while maintaining a balance between institutional goals, school policies, and sound educational practices.

Decision-making is widely recognized as the core of the administrative process and effective school leadership. School principals and teachers encounter numerous challenges in both instructional and administrative

functions, many of which hinder the achievement of school objectives. The ability of school administrators to make sound and timely decisions is therefore crucial in addressing these challenges and improving school effectiveness.

The concept of competence has become well established in organizational and educational literature. Spencer and Spencer (1993) define competencies as the underlying characteristics of an individual that are causally related to effective or superior performance in a given role. These competencies serve as indicators of an individual's ability to perform tasks according to established standards and criteria.

School administrators play a vital role in overseeing the daily operations of educational institutions. Their responsibilities include supervising teachers and staff, setting institutional goals, managing resources, communicating with stakeholders, and ensuring the effective delivery of instructional programs. In elementary and secondary schools, administrators—commonly referred to as principals—serve as a crucial link between students, teachers, parents, and the wider community. Their effectiveness largely depends on their leadership and decision-making competencies.

Abbott (1974), as cited by Aquino (2000), identified key decision-making competencies that contribute to effective school management. These include the ability to differentiate among types of information, determine

the appropriate amount and type of information needed for decisions, involve relevant stakeholders, establish priorities, and anticipate both intended and unintended consequences. These competencies are essential in minimizing negative outcomes and enhancing administrative effectiveness.

Research highlights the importance of participative decision-making in school administration. Teachers are central to the management and success of schools, and their involvement in decision-making fosters collaboration, commitment, and a shared sense of responsibility. Studies indicate that neglecting teacher participation often results in conflict, reduced morale, and resistance, which ultimately hinder the attainment of school goals. Conversely, involving teachers in decision-making promotes cooperation, dedication, and institutional support (Udo & Akpa, 2007).

Participative decision-making represents a shift from traditional top-down management to a more inclusive and collaborative leadership approach. Glew et al. (1995) describe it as an effort by higher-level administrators to provide subordinates with greater influence over organizational decisions. Jewell (1998) further emphasized that this approach helps prevent the “nobody asked” syndrome by encouraging employees to contribute ideas, better understand organizational issues, and more readily accept decisions that affect them.

Effective school management is grounded in the core managerial functions of planning, organizing, leading, and controlling. School

administrators are expected to act as instructional leaders, planners, supervisors, and facilitators of professional growth. A positive school climate characterized by openness, trust, and collaboration enhances teacher motivation, innovation, and instructional effectiveness (Short et al., 1991). In the context of Zamboanga del Sur, several socioeconomic and organizational factors influence school performance. The province is predominantly agro-based, with many families belonging to low- to lower-middle-income groups. These conditions contribute to challenges such as student absenteeism, limited instructional resources, frequent interruptions in classroom instruction, and inadequate educational facilities. Achievement test results in the Zamboanga del Sur Division from 2015 to 2018 consistently fell below the Department of Education's target proficiency level of 75 percent, indicating persistent concerns regarding academic performance.

Other factors affecting teaching and learning include non-instructional assignments given to teachers, limited availability of instructional materials and educational technology, leadership challenges, and insufficient participatory governance. These realities underscore the need for effective administrative management supported by strong decision-making competencies.

In view of these challenges, the researcher is motivated to conduct this study to determine the relationship between decision-making

competencies and administrative management in enhancing the teaching-learning process in the Zamboanga del Sur Division. This study aims to provide empirical evidence on how effective decision-making contributes to extraordinary school governance and improved educational outcomes.

Methods

The study employed the descriptive survey method of research. This method was considered appropriate because it aims to describe the existing conditions, practices, and phenomena as they occur in the present situation. The descriptive survey method focuses on gathering data to determine the current status of variables without manipulating them.

According to descriptive research literature, this method is used to describe the characteristics of a population or phenomenon being studied. It does not seek to explain cause-and-effect relationships but rather to present an accurate profile of situations, conditions, or perceptions through systematic data collection and analysis.

Research Instrument

The primary instrument used in this study was a self-developed questionnaire-checklist designed to gather data from school administrators

and teachers who served as respondents. The questionnaire-checklist consisted of two major parts:

- Part I focused on the decision-making competencies of school administrators for teaching–learning enhancement.
- Part II dealt with administrative management practices for teaching–learning enhancement.

Before the construction of the questionnaire-checklist, the researcher reviewed relevant books, master’s theses, dissertations, online resources, and other related literature to ensure content relevance and adequacy. The instrument was developed based on established theories and concepts related to decision-making competencies and administrative management. A draft questionnaire containing items aligned with the study’s objectives and sub-problems was then prepared.

Validation of the Research Instrument

After the questionnaire-checklist was developed, it was submitted to the researcher’s adviser for careful evaluation and refinement. Necessary revisions were made to ensure clarity, alignment with the research problems, and appropriateness of content. The revised instrument was then

presented to the Panel Members of the Graduate School of Southern Mindanao Colleges for further validation.

All comments and recommendations from the panel were incorporated into the instrument. The final version was subsequently submitted to the Dean of the Graduate School for approval. After approval, the questionnaire was converted into an online survey using Google Forms and subjected to a dry run among school administrators and teachers in the Pagadian City Division who were doctoral degree holders and not included in the actual respondents. Feedback from the dry run was reviewed, and further refinements were made before finalizing the questionnaire for actual data collection.

Data Gathering Procedure

Before the administration of the online questionnaire, the researcher formally requested permission from the Schools Division Superintendent of Zamboanga del Sur to conduct the study. Upon approval, the permission letter was attached to the online questionnaire and disseminated through school principals.

The respondents included school administrators and teachers who had served for three years or more in junior high schools within the Zamboanga del Sur Division. The researcher ensured that all respondents

completed the questionnaire fully by enabling the required response option in the Google Form, preventing unanswered items.

Sampling Design

The study utilized a purposive sampling technique. School administrators and teachers who met the criteria of having at least three years of service were deliberately selected as respondents due to their sufficient experience and direct involvement in administrative and instructional decision-making processes. This sampling design ensured that respondents possessed the relevant knowledge and experience necessary to provide valid and reliable data for the study.

Results

Decision-Making Competencies of School Administrators

The findings reveal that school administrators consistently demonstrated strong decision-making competencies in support of teaching–learning enhancement in the Zamboanga del Sur Division.

Skill in determining the type of decision was *always applied*, with an overall mean of 5.17. Administrators showed sound knowledge in making decisions related to administrative functions, instruction, personnel, and finance ($\bar{x} = 5.29$). They ensured that decisions were properly implemented and evaluated ($\bar{x} = 5.20$) and demonstrated awareness of the

implications of decisions in relation to information requirements and stakeholder involvement ($\bar{x} = 5.17$).

Skills in determining the type of information needed to reach a decision were *almost always applied*, with an average mean of 5.15. Administrators ensured that relevant information was gathered prior to decision-making ($\bar{x} = 5.24$), recognized the time-bound nature of the decision-making process ($\bar{x} = 5.18$), and collected sufficient information pertinent to identified problems ($\bar{x} = 5.17$).

Skills in determining the appropriate involvement of other stakeholders in decision-making were *always applied*, yielding the highest mean among decision-making variables (5.27). Administrators encouraged the participation of teachers, pupils, and parents ($\bar{x} = 5.52$), demonstrated the ability to determine appropriate levels of teacher involvement in decision implementation ($\bar{x} = 5.46$), and exercised dependable judgment regarding stakeholder participation ($\bar{x} = 5.40$).

Skills in establishing priorities for action were *almost always applied*, with an overall mean of 5.08. Administrators identified problems requiring immediate attention ($\bar{x} = 5.16$), demonstrated prudence in determining which activities should be implemented or deferred ($\bar{x} = 5.13$), and anticipated the consequences of immediate or delegated actions ($\bar{x} = 5.08$).

Skills in anticipating both intended and unintended consequences of decisions were likewise *almost always applied*, with an average mean of 5.14. Administrators exhibited competence in various decision-making situations ($\bar{x} = 5.23$), recognized that decisions may resolve one problem while creating another ($\bar{x} = 5.15$), and demonstrated the ability to estimate system outcomes ($\bar{x} = 5.13$).

Overall, the decision-making competencies yielded a grand mean of 5.16, indicating that these competencies were *almost always applied* by school administrators to enhance teaching–learning processes—an expression of extraordinary school governance.

Administrative Management Practices of School Administrators

With regard to administrative management, the results indicate that these practices were consistently and effectively applied by school administrators.

Organizing ranked first among the management functions, with an average mean of 5.38. Administrators analyzed the organizational structure of schools and offices ($\bar{x} = 5.48$), properly structured aims and objectives ($\bar{x} = 5.41$), and systematically arranged activities and programs ($\bar{x} = 5.37$).

Implementation ranked second, with an average mean of 5.37. Administrators carried out plans effectively ($\bar{x} = 5.51$), provided clear

implementation guides ($\bar{x} = 5.45$), and determined appropriate instructions prior to execution ($\bar{x} = 5.37$).

Planning was also *always applied*, with an average mean of 5.34. Administrators implemented plans on weekly, monthly, and yearly schedules ($\bar{x} = 5.47$), established programs for school improvement ($\bar{x} = 5.46$), and prepared master plans to guide classroom instruction ($\bar{x} = 5.38$). Coordination was *almost always applied*, with an average mean of 5.13. Administrators facilitated coordination through teacher guides ($\bar{x} = 5.24$), coordinated planning and organizing to address personnel limitations ($\bar{x} = 5.23$), and aligned school activities toward institutional objectives ($\bar{x} = 5.08$).

The overall mean for administrative management was 5.36, indicating that these practices were *always applied* in support of teaching–learning enhancement and extraordinary school governance.

Comparative and Inferential Analysis

The t-test analysis revealed no significant difference between the responses of school administrators and teachers regarding decision-making competencies (computed $t = 0.429$, tabular value = 2.306). Thus, the null hypothesis was accepted, indicating that both groups shared similar perceptions.

Similarly, the z -test results showed no significant difference between administrators' and teachers' responses concerning administrative management practices (computed $z = 0.95$, tabular value = 1.96). This further supports the acceptance of the null hypothesis.

Moreover, the correlation analysis between decision-making competencies and administrative management yielded a computed z value of 1.44, which is lower than the critical value of 1.96, indicating no significant relationship between the two variables in relation to teaching–learning enhancement.

Expression of Extraordinary School Governance

The findings affirm that decision-making competencies and administrative management practices are integral to effective school supervision. Although no significant relationship was found between the two variables, the consistent application of these competencies by school administrators contributes substantially to the enhancement of teaching–learning processes.

School administrators demonstrated competence in decision-making, inclusive governance, strategic planning, organization, implementation, and coordination. These practices foster collaboration, inspire teachers, and promote collective responsibility toward achieving institutional goals—thereby exemplifying an expression of extraordinary school governance.

Discussions

1. The skill in determining the appropriate type of decision was consistently applied by school administrators. They demonstrated competence in making decisions related to administrative operations, instructional supervision, personnel management, and financial concerns. Administrators also ensured that decisions were properly implemented and evaluated, showing awareness of their implications in terms of information requirements and stakeholder involvement.
2. The skill in identifying the type of information needed to arrive at a sound decision was almost always applied by school administrators. They ensured that relevant and sufficient information was gathered before decision-making, recognized the time-bound nature of the process, and collected data necessary to address identified concerns effectively.
3. The skill in determining the appropriate involvement of stakeholders in decision-making was consistently applied by school administrators. They encouraged the participation of teachers, learners, and parents, and demonstrated sound judgment in determining the level and nature of stakeholder involvement during decision implementation.

4. The skill in establishing priorities for action was almost always applied by school administrators. They identified issues requiring immediate attention, exercised prudence in determining which activities should be implemented or deferred, and demonstrated the ability to anticipate the consequences of immediate or delegated actions.
5. The skill in anticipating both intended and unintended consequences of decisions was almost always applied by school administrators. They displayed expertise across various decision-making situations and recognized that decisions may resolve certain issues while potentially creating new challenges, reflecting an understanding of system-wide outcomes.
6. Planning was consistently applied by school administrators through the preparation and implementation of weekly, monthly, and annual plans. They developed programs aimed at improving school operations and prepared master plans to guide teachers in formulating classroom-based plans and programs.
7. Organizing was consistently applied by school administrators by analyzing the school's organizational structure, aligning objectives with strategic approaches, and systematically arranging activities and programs to ensure efficiency.

8. Implementation was consistently applied by school administrators through the execution of approved plans and programs. Clear implementation guidelines and appropriate instructions were provided prior to actual implementation.
9. Coordination was almost always applied by school administrators by facilitating collaborative planning and organizing efforts, addressing personnel limitations, and ensuring that all school activities contributed to the achievement of institutional goals.
10. Overall, findings on the five decision-making competencies indicate that determining stakeholder involvement, identifying the type of decision, identifying necessary information, anticipating decision consequences, and establishing priorities for action were almost always applied by school administrators to enhance teaching–learning processes in the Zamboanga del Sur Division, reflecting extraordinary school governance.
11. In terms of administrative management, the four variables—planning, organizing, implementing, and coordinating—were found to be consistently applied by school administrators to support teaching–learning enhancement.
12. The test of significant difference between the responses of school administrators and teachers regarding decision-making competencies revealed that the computed t -value was lower than the

critical value. Thus, the null hypothesis was accepted, indicating no significant difference in the perceptions of the two respondent groups.

13. The test of significant difference between the responses of school administrators and teachers regarding administrative management showed that the computed z -value was lower than the critical value. Therefore, the null hypothesis was accepted, indicating no significant difference between the responses of the two groups.
14. The test of the relationship between decision-making competencies and administrative management for teaching–learning enhancement revealed that the computed z -value was lower than the critical value, indicating that there was no significant relationship between the two variables.

TECHNICAL VOCATIONAL LIVELIHOOD
CLASSROOM SOCIAL ENVIRONMENT AND THE
LEVEL OF CLASSROOM PARTICIPATION AMONG
SENIOR HIGH SCHOOL STUDENTS OF QUALITY
LEARNING CIRCLE 4 IN ZAMBOANGA DEL SUR

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Modified Title: Technical-Vocational Livelihood Classroom Social
Environment and the Level of Classroom Participation among Senior High
School Students of Quality Learning Circle 4 in Zamboanga del Sur

Abstract

This study examined the relationship between the Technical-Vocational Livelihood (TVL) classroom social environment and the level of classroom participation among Senior High School students in Quality Learning Circle 4 (QUALCI 4), Zamboanga del Sur. Specifically, it assessed students' perceptions of the classroom social environment in terms

of teacher–student relationship, classroom civility, student–student relationship, and TVL facilities, as well as their level of procedural and substantive classroom participation. A descriptive-quantitative research design was employed involving 494 TVL students selected through stratified random sampling during the School Year 2018–2019. Data were gathered using researcher-developed questionnaires validated by experts and tested for reliability. Statistical tools such as frequency, percentage, mean, standard deviation, Pearson product-moment correlation, t-test, and one-way ANOVA were used to analyze the data at the 0.05 level of significance. Results revealed that respondents perceived the classroom social environment as generally positive ($M = 3.25$). Classroom participation was also found to be high, with procedural participation rated as Participative ($M = 3.31$) and substantive participation as Very Participative ($M = 3.24$). Significant positive relationships were found between the classroom social environment and both procedural and substantive participation, except for the relationship between TVL facilities and substantive participation, which was not significant. Significant differences were likewise observed in classroom social environment and participation when respondents were grouped according to selected demographic variables. The findings suggest that positive teacher–student relationships, respectful peer interactions, and a supportive classroom climate significantly enhance students' participation in TVL classes. The

study recommends strengthening classroom social relationships and fostering an inclusive learning environment to further improve student engagement and participation.

Keywords: classroom social environment, classroom participation, Technical-Vocational Livelihood (TVL), Senior High School, teacher–student relationship, classroom civility, Zamboanga del Sur

Introduction

The classroom environment plays a vital role in shaping students' learning experiences and academic success. In particular, the social atmosphere within the classroom influences students' motivation, behavior, and engagement in learning activities. A positive classroom social environment, characterized by supportive teacher-student relationships, respectful peer interaction, and a conducive learning climate, encourages students to participate actively in class. This is especially important in Technical-Vocational Livelihood (TVL) classrooms, where learning involves hands-on activities, collaboration, and practical application of skills.

Classroom participation is a key indicator of student engagement and learning effectiveness. It reflects the extent to which students are

involved in class discussions, group work, and classroom tasks. Students who actively participate in class are more likely to develop confidence, critical thinking skills, and a deeper understanding of the subject matter. Conversely, limited participation may lead to poor academic performance, lack of motivation, and reduced skill development.

In Quality Learning Circle 4 (QUALCI 4) in Zamboanga del Sur, senior high school students enrolled in TVL programs are expected to develop technical competencies and prepare for future employment. However, differences in classroom participation have been observed, which may be influenced by the classroom social environment. Factors such as teacher-student interaction, classroom civility, peer relationships, and the availability of TVL facilities may affect students' willingness to engage in classroom activities.

This study aims to assess the TVL classroom social environment and the level of classroom participation among senior high school students in Quality Learning Circle 4, Zamboanga del Sur. By examining the relationship between the classroom social environment and students' participation, the study seeks to provide insights that can help improve instructional strategies, enhance classroom relationships, and promote a more engaging learning environment for TVL students.

Methods

Research Design

This study utilized a descriptive–quantitative research design to examine the Technical-Vocational Livelihood (TVL) classroom social environment and the level of classroom participation among Senior High School students of Quality Learning Circle 4 in Zamboanga del Sur. The design was appropriate for describing existing conditions and determining relationships between variables without manipulation.

Research Locale

The study was conducted in selected Senior High Schools under Quality Learning Circle 4 (QUALCI 4) in Zamboanga del Sur during School Year 2018–2019. These schools offer the Technical-Vocational Livelihood track, specifically the Industrial Arts strand.

Respondents and Sampling Technique

The respondents consisted of 494 Senior High School students enrolled in the TVL track from selected schools in QUALCI 4. A stratified random sampling technique was employed to ensure proportional representation based on school and TVL strand. Students of varying age, sex, ethnicity, and academic background were included in the study.

Research Instruments

Two researcher-developed survey questionnaires were used:

1. Classroom Social Environment Questionnaire, which measured students' perceptions of the classroom social environment in terms of:
 - teacher–student relationship,
 - classroom civility,
 - student–student relationship, and
 - TVL facilities.
2. Classroom Participation Questionnaire, which assessed students' level of classroom participation in terms of:
 - procedural participation and
 - substantive participation.

Both instruments used a five-point Likert scale ranging from *1 (Strongly Disagree)* to *5 (Strongly Agree)*.

Validity and Reliability

The instruments were evaluated by experts in education and research to establish content validity. A pilot test was conducted among students not included in the final sample, and the reliability of the

questionnaires was determined using Cronbach's alpha, which yielded acceptable reliability coefficients.

Data Collection Procedure

Approval to conduct the study was obtained from the school authorities. The questionnaires were administered personally by the researcher with the assistance of class advisers. Respondents were informed of the purpose of the study and assured of confidentiality. Completed questionnaires were retrieved, reviewed, and prepared for statistical analysis.

Statistical Treatment of Data

The data were analyzed using the following statistical tools:

- Frequency and Percentage – to describe respondents' demographic profile;
- Mean and Standard Deviation – to determine the levels of classroom social environment and classroom participation;
- Pearson Product–Moment Correlation (r) – to determine the relationship between the two main variables;
- t-test and One-Way ANOVA – to determine significant differences when data were grouped according to profile variables.

All statistical analyses were conducted at a 0.05 level of significance.

Results

1. The study involved 494 Senior High School students enrolled in the Technical-Vocational Livelihood (TVL) track from schools under Quality Learning Circle 4 in Zamboanga del Sur. Most respondents were 16–17 years old, predominantly male, largely of Visayan ethnicity, and mainly enrolled in the Industrial Arts strand.
2. The classroom social environment, assessed in terms of teacher–student relationship, classroom civility, student–student relationship, and TVL facilities, yielded an overall mean of 3.25, interpreted as Agree, indicating a generally positive classroom social environment.
3. The level of classroom participation was high. Procedural participation obtained an overall mean of 3.31, described as Participative, while substantive participation recorded a mean of 3.24, described as Very Participative.
4. A significant relationship was found between the classroom social environment and procedural participation across all components, including teacher–student relationship, classroom civility, student–student relationship, and TVL facilities.

5. A significant relationship was also observed between the classroom social environment and substantive participation in terms of teacher–student relationship, classroom civility, and student–student relationship. However, no significant relationship was found between substantive participation and TVL facilities.
6. There was a significant difference in the classroom social environment when respondents were grouped according to profile variables across all components. In contrast, no significant difference was noted in TVL facilities when grouped according to sex.
7. Significant differences were found in both procedural and substantive classroom participation when respondents were grouped according to their profile variables.

Discussions

The findings of the study reveal that the classroom social environment in Technical-Vocational Livelihood (TVL) classes under Quality Learning Circle 4 (QUALCI 4) is generally positive. Students perceived their learning environment as supportive, characterized by respectful teacher–student relationships, harmonious peer interactions, and a generally conducive classroom atmosphere. This result is significant because the

nature of TVL education requires collaborative and hands-on learning, where students rely heavily on interactions with teachers and peers to complete tasks and develop competencies. A positive classroom social environment, therefore, serves as a foundation for effective learning and engagement.

The study further indicates that students in TVL classes exhibit high levels of classroom participation. Procedural participation—such as following classroom rules, completing assigned tasks, and adhering to routines—was found to be participative. Substantive participation, which includes meaningful engagement such as contributing ideas, asking questions, and participating in discussions, was rated as very participative. This suggests that students are not only compliant with classroom procedures but are also actively involved in learning activities that require critical thinking and collaboration. This level of participation is essential in TVL education, where practical skills are developed through active involvement and repeated practice.

A key finding of the study is the significant relationship between the classroom social environment and classroom participation. Specifically, the teacher-student relationship, classroom civility, and student-student relationship were strongly linked to both procedural and substantive participation. These results imply that students are more likely to participate when they feel supported, respected, and valued by their teachers and

classmates. When teachers foster an atmosphere of trust and encouragement, students tend to be more confident in expressing their thoughts and taking part in learning activities. Similarly, positive peer relationships promote cooperation and group engagement, which enhances participation.

Interestingly, TVL facilities showed a significant relationship with procedural participation but not with substantive participation. This suggests that while the availability of facilities and equipment supports students' ability to perform routine tasks and complete practical activities, it does not necessarily encourage deeper engagement in discussions or critical thinking.

This highlights that while physical resources are important, they are not the primary factor that drives meaningful participation. Instead, interpersonal dynamics and classroom culture play a more influential role in encouraging students to participate actively and confidently.

Moreover, the study found significant differences in the classroom social environment and classroom participation when respondents were grouped according to profile variables such as age, sex, ethnicity, and strand. This indicates that student demographics may influence their perception of the classroom environment and their level of engagement. For instance, differences in age and maturity may affect students' confidence and willingness to participate, while cultural and social backgrounds may

shape how students interact with teachers and peers. These differences emphasize the need for educators to consider individual and cultural factors when creating a supportive learning environment.

Overall, the findings highlight the critical role of a positive classroom social environment in promoting student participation in TVL classes. Fostering strong interpersonal relationships, mutual respect, and a supportive classroom culture can significantly enhance students' engagement and involvement in learning activities. Therefore, school administrators and teachers should prioritize creating an environment that encourages collaboration, respect, and active participation to improve learning outcomes and student performance in TVL education.

FINANCIAL LITERACY AND FINANCIAL MANAGEMENT PRACTICES AMONG PUBLIC ELEMENTARY SCHOOL TEACHERS IN SELECTED SCHOOLS IN THE SCHOOLS DIVISION OF NUEVA VIZCAYA

Professor Regino L. Galate, Teddy A. Buduan, Larry D. Somera,
Maricel B. Belingon, and Joy D. Balawa

Modified Title: Financial Literacy and Its Relationship to Financial
Management Practices among Public Elementary School Teachers in the
Schools Division of Nueva Vizcaya

Abstract

This study examined the level of financial literacy and its relationship to financial management practices among public elementary school teachers in selected schools in the Schools Division of Nueva Vizcaya. Specifically, it assessed teachers' financial literacy in terms of saving, budgeting, and investment; financial behavior; financial planning; and the influence of selected socio-demographic factors on these variables.

The study employed a descriptive-comparative and descriptive-correlational research design involving 64 permanent public elementary school teachers selected through purposive sampling. Data were gathered using a structured questionnaire and analyzed using descriptive statistics, correlation analysis, and tests of significant difference. Findings revealed that the respondents demonstrated an adequate level of financial literacy, characterized by strong saving practices and moderate budgeting skills but limited knowledge and participation in investment activities. Teachers also exhibited generally responsible financial behavior, particularly in spending and bill payment, although long-term financial planning remained insufficient. Significant positive relationships were found between financial literacy and financial behavior, financial literacy and financial planning, and financial behavior and financial planning. Moreover, age, gender, and civil status significantly influenced selected aspects of financial literacy and financial behavior, while household size showed no significant effect. The study concludes that improving teachers' financial literacy, particularly in investment and long-term financial planning, can enhance responsible financial management practices and strengthen their overall financial well-being.

Keywords: financial literacy, financial management practices, financial behavior, financial planning, public elementary school teachers, socio-demographic factors, savings, budgeting, investment, Nueva Vizcaya

Introduction

Financial Literacy and Financial Management Practices

In the context of rapid economic development and increasing financial complexity, financial literacy has become an essential life skill that significantly influences both individual and global financial well-being. The Organization for Economic Co-operation and Development (OECD) recognizes financial literacy as a core competence in the 21st century, emphasizing its importance in navigating modern financial systems and making sound financial decisions (OECD, n.d.).

Financial literacy enables individuals to make informed, effective decisions about managing their financial resources. It involves knowledge and understanding of fundamental financial concepts such as budgeting, saving, investing, and debt management. As financial challenges continue to intensify, financial literacy is increasingly regarded as a vital competency, particularly among educators who serve as role models and financial influencers within their communities.

Elementary school teachers play a crucial role in shaping the knowledge, values, and behaviors of young learners. Beyond their professional responsibilities, teachers face personal financial obligations that directly affect their well-being and job performance. Effective financial management is essential in ensuring financial stability and security; however, many teachers encounter financial difficulties due to limited financial knowledge, inadequate budgeting skills, and insufficient savings or investment practices. Given their significant influence on future generations, understanding teachers' financial literacy and financial management practices is vital in fostering a financially responsible and resilient teaching workforce.

This study focuses on the financial literacy and financial management practices of public elementary school teachers in selected schools within the Schools Division of Nueva Vizcaya. By examining teachers' financial knowledge and financial behaviors, the study seeks to identify existing gaps and provide recommendations to enhance financial literacy and financial management within the educational sector.

Understanding Financial Literacy and Financial Management

Bungabong and Rosalejos (2024) emphasized that financial literacy and financial management are fundamental components of personal finance that significantly influence an individual's financial well-being. In today's

complex and dynamic economic environment, these competencies are indispensable for making informed decisions, achieving financial goals, and ensuring long-term financial security. Despite increased awareness, many individuals continue to lack sufficient financial education and access to financial resources.

Financial literacy, also referred to as financial capability, is defined as an individual's ability to manage financial resources effectively to achieve sustained financial security. It consists of two key components: financial understanding and application (Abaya et al., 2021). Sanderson (2015, as cited by Delmo et al., 2023) described financial literacy as the ability to apply financial knowledge and skills to make appropriate financial decisions and manage financial resources efficiently. Similarly, Sujaini (2021, as cited by Delmo et al., 2023) defined financial literacy as the competence to understand and utilize financial skills such as budgeting, saving, and financial management. Financially literate individuals are more likely to achieve self-sufficiency and financial security.

Dwiastanti (2015, as cited by Delmo et al., 2023) highlighted the growing importance of financial literacy research, emphasizing that individuals must possess adequate financial knowledge to manage their finances effectively and attain financial prosperity. Financial literacy positively influences financial capability and personal financial behavior (Akca et al., 2018; Te'eni-Harari, 2016, as cited by Rapina, 2023). The

development of financial literacy is a continuous process that requires sustained effort from individuals and support from educational institutions and the broader social environment (Çera et al., 2021, as cited by Rapina, 2023).

Individuals with higher levels of financial literacy are more likely to demonstrate responsible financial behavior, particularly in long-term financial decisions such as home ownership and retirement planning (Blue et al., 2014, as cited by Rapina, 2023). Several studies confirm that as financial literacy increases, individuals' capacity to manage financial resources effectively also improves (Akca et al., 2018; Arofah et al., 2018, as cited by Rapina, 2023).

Financial Management Practices

Financial management is recognized by the World Bank Group as a key component of good governance and a critical factor in achieving sustainable development goals, including poverty reduction and shared prosperity (World Bank Group, n.d.). Pandey (2025) defined financial management as the systematic process of planning, organizing, directing, and controlling financial resources to achieve goals efficiently and effectively. It includes budgeting, forecasting, cash flow management, investment decisions, risk management, and financial reporting, with the primary objective of ensuring financial sustainability and stability.

Financial management practices refer to a combination of financial knowledge, skills, attitudes, and behaviors that guide individuals in making informed financial decisions (Borres, 2023). Hussain and Sajjad (2018, as cited by Borres, 2023) described these practices as essential elements contributing to financial well-being, including budgeting, saving, borrowing, investing, and spending. Engagement in these practices is influenced by perceived importance, individual capacity, and available financial opportunities (Ecija, 2020).

Moreover, financial management practices encompass activities such as capital budgeting, accounting, cash flow management, financial planning and control, and working capital management (Nkundabanyanga et al., 2017, as cited by Mang'ana, 2023). In a personal or household context, financial management includes income and expenditure management, saving, investing, debt management, and long-term financial planning (Low, 2024, as cited by Prakoso & Apriliani, 2024).

Budgeting is a fundamental financial management practice that facilitates planning, communication, and control of financial resources. It enables individuals to allocate resources effectively to achieve financial goals (Siegel et al., 2014, as cited by Alcantara, 2024). Budgeting translates financial plans into measurable and actionable steps, allowing individuals to monitor performance and make necessary adjustments (Hanninen, 2014, as cited by Alcantara, 2024).

Saving is another essential component of effective financial management, as it promotes financial security and preparedness for unforeseen circumstances. Regular saving helps individuals build financial reserves, reduce reliance on debt, and support long-term objectives such as retirement and asset acquisition. When combined with budgeting, saving contributes significantly to long-term financial stability and well-being (Prakoso & Apriliani, 2024).

Investment involves allocating resources with the expectation of generating future returns (Nguyen et al., 2020, as cited by Borres, 2023). Early investment enhances the growth potential of savings through compounding, allowing assets to accumulate over time. Investing early in diversified portfolios supports wealth accumulation and helps mitigate risks associated with inflation and market volatility (Yoganandham, 2025).

Influence of Socio-Demographic Factors on Financial Literacy and Financial Management

Several studies have established that financial literacy positively influences financial behavior (Ali et al., 2016, as cited by Murdiono et al., 2023). Demographic factors also play a significant role in shaping financial management behavior. Murdiono et al. (2023) noted that characteristics such as age, gender, and income significantly affect financial decision-

making. Similarly, Humaidi et al. (2020, as cited by Murdiono et al., 2023) found that demographic variables influence financial management behavior.

Borres (2023) reported that financial literacy is strongly associated with socio-demographic characteristics and family structure. Age, health status, household size, and income level were identified as key determinants of financial literacy (Mark Taylor, 2018, as cited by Borres, 2023). Delmo et al. (2023) explained that demographic segmentation is a research approach that categorizes populations based on variables such as gender, age, income, marital status, and occupation. Gender differences influence financial preferences and behaviors (Zafer & Aslihan, 2012, as cited by Delmo et al., 2023), while age-based categorization reflects varying financial needs and priorities across life stages. Within the teaching profession, a substantial proportion of educators are married and tenured, which further shapes their financial responsibilities and management practices.

Keywords: Financial Literacy; Financial Behavior; Financial Management Practices; Financial Planning; Public Elementary School Teachers; Socio-Demographic Factors; Savings, Budgeting, and Investment; Nueva Vizcaya

METHODS

Research Design

This study employed a descriptive–comparative and descriptive–correlational research design. The descriptive–comparative approach was used to determine the differences in financial behavior and financial management practices of public elementary school teachers when grouped according to their profile variables. The descriptive–correlational design was utilized to examine the degree of relationship between the respondents’ financial behavior and their financial management practices.

Locale of the Study and Respondents

The study was conducted in selected public elementary schools within the Schools Division of Nueva Vizcaya, namely: Bintawan South Elementary School, Ernesto “Lapan” Pale Elementary School, Felix–Juana Brawner Community School, Governor Juan Manzano Elementary School, Nagbitin Elementary School, Ocapon Elementary School, Sawmill Elementary School, Villa Par-Pale Elementary School, and Governor Alfonso Castañeda Elementary School.

A total of 64 permanent public elementary school teachers served as respondents of the study. The respondents were selected using purposive sampling, with years of teaching experience and length of service in their

respective schools as the primary criteria for inclusion. Teachers with non-permanent employment status and those with less than one year of teaching experience were excluded from the study to ensure the reliability and relevance of the data gathered.

Distribution of Respondents

Distribution of Respondents

School	Sample (n)	Percentage (%)
Governor Alfonso Castañeda ES	9	
Bintawan South ES	11	
Ernesto “Lapan” Pale ES	9	
Felix–Juana Brawner Community School	7	
Governor Juan Manzano ES	4	
Nagbitin ES	7	
Ocapon ES	7	
Sawmill ES	7	
Villa Par-Pale ES	3	
Total	64	

Results

This study assessed the levels of financial literacy, financial behavior, and financial planning of selected public elementary school teachers in Nueva Vizcaya and examined the relationships among these variables as well as their association with selected demographic factors. The major findings are summarized as follows:

1. Level of Financial Literacy

- 1.1. Teachers demonstrated a good level of saving practices ($M = 3.53$), frequently recognizing the importance of saving and engaging in basic saving behaviors.
- 1.2. Budgeting skills were found to be adequate ($M = 3.33$), indicating moderate awareness of income and expenditures but inconsistent implementation of structured budgeting and saving practices.
- 1.3. Investment practices were rated low ($M = 2.28$), as
- 1.4. most teachers seldom participated in insurance, real estate, or stock market investments.
- 1.5. Overall, the level of financial literacy was adequate ($M = 3.01$), characterized by strength in saving, moderate competence in budgeting, and weakness

investment knowledge and practices.

2. Financial Behavior

- 2.1. Teachers exhibited good purchase behavior ($M = 3.99$), prioritizing necessities, comparing prices, and practicing responsible spending habits.
- 2.2. Promptness and timeliness of bill payments were also rated good ($M = 3.83$), although some respondents reported settling payments only at or near due dates.
- 2.3. Long-term financial goals were assessed as adequate ($M = 3.47$), reflecting awareness of future financial needs but limited engagement in systematic saving and investment activities.
- 2.4. The incidence of debt was rated low ($M = 2.29$), suggesting minimal reliance on loans, though some teachers experienced challenges in managing existing obligations.
- 2.5. Overall, teachers' financial behavior was adequate ($M = 3.39$).

3. Financial Planning for Future Expenses

- 3.1. Teachers generally demonstrated awareness of future financial needs, particularly retirement (77.6%) and

health care (72.4%). However, this awareness did not consistently translate into concrete financial planning and fund allocation.

- 3.2. For example, while 77.6% of teachers acknowledged the need for retirement preparation, only 65.5% reported having a plan, and merely 43.1% had allocated funds for this purpose.
- 3.3. The lowest levels of awareness and preparedness were observed for discretionary future expenses, such as weddings and vehicle purchases.

4. Relationship Between Variables

- 4.1. A moderate positive correlation was found between financial literacy and financial behavior ($r = 0.596$, $p < .01$), indicating that higher financial literacy is associated with better financial practices.
- 4.2. A moderate positive correlation also existed between financial literacy and financial planning ($r_{pb} = 0.364$, $p < .01$), suggesting that increased financial knowledge supports improved planning capabilities.

4.3. Similarly, financial behavior and financial planning were positively correlated ($r_{pb} = 0.326$, $p < .05$), implying that responsible financial practices enhance readiness for future financial needs.

5. Influence of Demographic Factors

- a. Age showed a significant positive correlation with financial literacy ($r = 0.182$, $p < .05$) and financial behavior ($r = 0.210$, $p < .05$), indicating that older teachers demonstrated slightly higher levels of literacy and better financial practices. No significant relationship was found between age and financial planning.
- b. Household size did not significantly influence financial literacy, financial behavior, or financial planning.
- c. Gender differences were significant only in financial literacy ($p = .038$), with male teachers scoring higher than female teachers. No significant gender differences were found in financial behavior, while financial planning showed only marginal variation.
- d. Civil status significantly affected financial literacy ($p = .007$) and financial behavior ($p = .038$), with married teachers performing better than their single or widowed

counterparts. No significant differences were observed in financial planning.

Discussions

Public elementary school teachers exhibit an adequate level of financial literacy, with strong saving practices but limited knowledge and participation in investment activities. This generally cautious and risk-averse financial orientation may be attributed to constrained income levels and minimal exposure to investment opportunities.

□ Teachers demonstrate moderately sound financial behavior, particularly in responsible spending and timely payment of financial obligations. However, their long-term financial strategies—especially in investment and future-oriented planning—remain insufficiently developed.

□ While teachers are aware of the importance of financial planning for retirement, health care, and educational needs, this awareness is not consistently translated into concrete actions. Many struggle to convert financial knowledge into regular savings and structured financial plans.

□ Financial literacy has a significant influence on financial behavior and financial planning, and financial behavior likewise contributes meaningfully to planning outcomes. These findings indicate that financial knowledge and practices operate synergistically in enhancing overall financial preparedness.

□ Demographic variables influence financial outcomes selectively. Age and civil status positively affect financial literacy and financial behavior, whereas gender differences are evident only in financial literacy. Household size does not significantly affect financial outcomes, suggesting that individual discipline and financial knowledge play a more critical role than family composition.

USE OF ELECTRICAL WIRING BOARD IN IMPROVING THE ACADEMIC PERFORMANCE OF GRADE 8 IN ELECTRICAL INSTALLATION AND MAINTENANCE

LARRY DINU SOMERA

Modified Title: Use of an Electrical Wiring Board to Improve the
Academic Performance of Grade 8 Students in Electrical Installation and
Maintenance

Abstract

This study determined the effectiveness of a teacher-developed electrical wiring board (circuit trainer) in improving the academic performance of Grade 8 students in Electrical Installation and Maintenance (EIM) at Dupax Del Sur National High School. Specifically, it compared the academic performance of students taught using the electrical wiring board with those taught through the conventional method of instruction. The study employed a quantitative approach using a quasi-experimental pretest–posttest control group design involving 60 Grade 8 students randomly

assigned to either the control or experimental group. A researcher-developed and validated 75-item achievement test was administered before and after the intervention to measure students' learning outcomes. Data were analyzed using mean, standard deviation, independent samples t-test, and paired samples t-test at the 0.05 level of significance. Results revealed that both groups initially demonstrated low mastery of EIM competencies, with no significant difference in their pretest scores. Following the intervention, both groups improved to an average level of mastery; however, the experimental group obtained significantly higher posttest scores than the control group. Significant learning gains were likewise observed between the pretest and posttest scores of both groups, with greater improvement recorded among students exposed to the electrical wiring board. The findings demonstrate that the use of the electrical wiring board is an effective instructional innovation that enhances students' academic performance, promotes hands-on learning, and improves the mastery of practical competencies in Electrical Installation and Maintenance.

Keywords: electrical wiring board, circuit trainer, Electrical Installation and Maintenance (EIM), academic performance, Grade 8 students, instructional innovation, hands-on learning, Technology and Livelihood Education, quasi-experimental design, technical-vocational education

Introduction

During the past four years of teaching at Dupax Del Sur National High School, the researcher encountered numerous challenges, including adapting to new systems, instructional materials, and teaching strategies, as well as gaining an understanding of classroom structure and regimentation through professional experience. Among these challenges, identifying appropriate instructional materials that effectively facilitate teaching and learning in the classroom proved to be the most difficult. Through a process of trial and error, the researcher gradually developed strategies to enhance classroom management, minimize disruptions, and improve student engagement. These experiences led to the development of an instructional innovation known as the **Electrical Wiring Board**, also called a **Circuit Trainer**.

The Philippine government, through the establishment of the **Technical Education and Skills Development Authority (TESDA)**, has prioritized workforce development as a key driver of economic growth. TESDA plays a significant role in advancing **Technical Vocational Education and Training (TVET)** by encouraging research-based innovations that improve instructional delivery. Research findings and recommendations serve as foundations for policy formulation aimed at

enhancing students' academic performance while developing competent, productive, and globally competitive workers (TESDA, 2010).

In support of this mandate, TESDA formulated the **National Technical Education and Skills Development Plan (NTESDP)**, which envisions a vibrant and high-quality TVET system that promotes decent work and sustainable inclusive growth. The plan is guided by three strategic thrusts: **TVET for Global Competitiveness and Workforce Readiness**, **TVET for Social Equity and Workforce Inclusion**, and **TVET for Poverty Reduction** (TESDA, 2010).

The first thrust, TVET for Global Competitiveness, emphasizes the development of skills, knowledge, and technologies essential for industrial productivity. Productivity is widely recognized as a cornerstone of sustainable economic development and wealth generation (TESDA, 2011). To remain globally competitive, TVET institutions must adopt internationally recognized standards, continuously integrate emerging technologies and innovations, align training programs with industry demands, and strengthen partnerships between public and private sectors. These efforts aim to equip learners with adaptable and industry-relevant skills responsive to technological advancements (TESDA, 2011).

Global competitiveness has become increasingly urgent due to trade liberalization and investment facilitation, which allow the free movement of goods, services, and labor across national borders. In response, TESDA

promotes structured training delivery systems implemented through schools, training centers, and enterprises. These systems cater to wage and self-employed workers, future industry practitioners, educators, vocational and higher education graduates, and workers in emerging industries. Core initiatives include competency standards development, training regulation, program accreditation, competency assessment and certification, scholarship programs, skills competitions, and the establishment of centers of excellence.

The second thrust, TVET for Social Equity, underscores inclusive workforce development by addressing the needs of marginalized sectors, such as informal workers, farmers, fisherfolk, indigenous peoples, overseas Filipino workers, women, persons with disabilities, and other socially disadvantaged groups. From 2018 to 2022, TESDA identified three key employment generators: rapidly growing industries such as tourism, construction, and transportation; traditional sectors with large employment bases but slow growth such as agriculture and manufacturing; and high-growth service industries including health, wellness, and social services.

Despite these initiatives, TESDA identified several challenges confronting the TVET sector. These include rapid technological advancements associated with the **Fourth Industrial Revolution (4IR)**, evolving workforce competency requirements, and the persistent issue of low employability among some TVET graduates. Additionally, the country

faces a significant labor outflow, with millions of skilled Filipino workers employed overseas, resulting in a depletion of highly skilled labor domestically (TESDA, 2013). To address these challenges, TESDA established five strategic objectives focused on improving skills development, employability, innovation, and workforce competitiveness.

TESDA emphasizes the need for research-driven, innovative, and adaptive instructional strategies to prepare learners for the demands of the modern labor market. As highlighted by TESDA Deputy Director General Rosanna Urdaneta, academic institutions, researchers, employers, and government agencies must collaborate to ensure global competitiveness. Research findings and recommendations are encouraged to inform policies and strengthen TVET delivery systems nationwide.

Similarly, the **Asia-Pacific Economic Cooperation (APEC)** recognizes that human capital is a critical resource for economic growth. APEC has expressed concern over the mismatch between industry skill requirements and the competencies produced by educational institutions. Studies conducted by APEC highlight the importance of aligning education and training programs with global industry standards to address skills shortages and workforce gaps.

At the school level, **Technology and Livelihood Education (TLE)** plays a crucial role in preparing learners for career pathways by equipping them with practical knowledge, skills, work values, and problem-solving

abilities, particularly in the field of Industrial Arts. The effective implementation of the **K to 12 Basic Education Curriculum** further emphasizes the development of 21st-century skills through relevant, experiential, and skills-based learning. However, the successful delivery of these objectives depends largely on the availability of appropriate instructional materials, tools, and equipment.

The researcher, currently employed as a **Teacher III** specializing in **Electrical Installation and Maintenance (EIM)**, has taught the subject for over ten years. Throughout this period, the researcher observed persistent difficulties in teaching practical skills due to the lack of adequate instructional materials. In response, the researcher designed and developed a **teacher-constructed electrical circuit trainer** to enhance instruction and assess its effectiveness in improving the academic performance of Grade 8 students at Dupax Del Sur National High School.

The electrical circuit trainer consists of a movable instructional board measuring approximately 198 centimeters in height and 87 centimeters in width. It includes multiple circuit panels equipped with light-emitting diode (LED) bulbs, a fluorescent lamp, single-pole switches, three-way switches, and convenience outlets. This equipment allows students to perform hands-on activities simulating real-life electrical installations.

The use of the electrical circuit trainer provides meaningful opportunities for experiential learning, supplementing traditional teaching methods such as board-and-chalk instruction. It enhances students' knowledge, skills, and attitudes by promoting active engagement, practical application, and skill mastery. Ultimately, this instructional innovation supports the development of globally competitive, skilled, and productive learners aligned with the goals of TESDA and the national education system.

Keywords: Electrical Wiring Board; Circuit Trainer; Electrical Installation and Maintenance (EIM); Academic Performance; Grade 8 Students; Technical Vocational Education and Training (TVET); Instructional Materials; Hands-on Learning; Skills Development; Technology and Livelihood Education (TLE)

Methods

Research Design

This study employed a quantitative research approach using a quasi-experimental pretest–posttest control group design. Quantitative research involves the collection and analysis of numerical data to describe, compare,

and explain phenomena through statistical procedures. This approach enables objective measurement of students' academic performance and the determination of the effectiveness of instructional interventions.

Two comparable groups—the control group and the experimental group—were utilized. Both groups were administered a pretest in Electrical Installation and Maintenance (EIM) to determine their entry-level competencies. The experimental group was taught using the teacher-constructed electrical wiring board (circuit trainer), while the control group was exposed to the conventional method of teaching. After the intervention, a posttest parallel to the pretest was administered to both groups to measure learning gains.

The research design is illustrated as follows:

R O₁ X O₂

R O₃ O₄

Where:

R – Random assignment

X – Treatment (use of the electrical wiring board)

O₁ – Experimental group pretest

O₂ – Experimental group posttest

O₃ – Control group pretest

O₄ – Control group posttest

Research Method

The study utilized the quasi-experimental method, which allows the examination of cause-and-effect relationships under controlled conditions without full randomization. The independent variable was the use of the electrical wiring board, while the dependent variable was the academic performance of Grade 8 students in EIM. Both groups underwent pretesting to establish comparability. The control group received instruction through the traditional lecture and demonstration method, while the experimental group received instruction through lectures integrated with hands-on activities using the circuit trainer. The intervention was implemented over a specified instructional period.

Considering the educational nature of the study, the level of significance was set at 5%, which is standard in educational and social science research.

Subjects / Respondents of the Study

The respondents of the study consisted of 60 Grade 8 students, representing 85.99% of the total population of 70 students from two sections at Dupax Del Sur National High School. From the first section with 33 students, 30 students (90.90%) were selected, while 30 students (81.08%) were chosen from the second section with 37 students.

Respondents were selected based on their most recent achievement test results to ensure comparable entry-level performance. Using the fishbowl technique, 30 students were randomly assigned to the control group and 30 to the experimental group. A lottery method determined group assignment.

Table 1. Frequency and Percentage Distribution of Respondents

Group	Population Sample Percentage		
Control	33	30	90.90%
Experimental	37	30	81.08%
Total	70	60	85.99%

Research Instruments

Electrical Installation and Maintenance Achievement Test

The primary data-gathering instrument was a researcher-developed 75-item multiple-choice test based on the K to 12 EIM Curriculum Guide. The distribution of test items reflected the relative importance of each competency area as specified in the curriculum.

The test instrument underwent face and content validation by five external experts and three panelists in Industrial Technology Education

from both secondary and tertiary institutions. The validators assessed clarity, content relevance, appropriateness of language, and alignment with learning objectives.

Pretest

The pretest consisted of 75 multiple-choice items covering selected EIM competencies and was administered before the experimental intervention to determine baseline knowledge and skills.

Posttest

The posttest consisted of 75 parallel items paraphrased from the pretest and was administered after the intervention to assess learning gains.

Electrical Wiring Board (Circuit Trainer)

The electrical wiring board was designed and constructed by the researcher in accordance with the EIM curriculum. It was used as the primary instructional tool for the experimental group to facilitate hands-on and experiential learning.

Data Gathering Procedure

Permission to conduct the study was secured from the school principal of Dupax Del Sur National High School. After approval, the researcher prepared the table of specifications and developed the research

instruments, which were subsequently validated by experts. The respondents were then identified and assigned to either the control or experimental group. Both groups were administered the pretest to determine comparability. Upon establishing similar pretest performance, the experimental intervention was conducted.

The control group was taught using the conventional method of instruction, while the experimental group was taught using the electrical wiring board. After the intervention, the posttest was administered to both groups. The collected data were subjected to statistical analysis to answer the research questions.

Statistical Tools Used

The following statistical tools were employed using Microsoft Excel:

- Mean and Standard Deviation – used to describe students' performance in the pretest and posttest.
- Independent Samples t-test – used to determine the significant difference between the mean scores of the control and experimental groups.
- Paired Samples t-test – used to determine the significant difference between pretest and posttest mean scores within each group.

All statistical analyses were interpreted at the 5% level of significance.

Results

□ The pretest results revealed that both the control group ($M = 18.64$) and the experimental group ($M = 19.40$) demonstrated low mastery in Electrical Installation and Maintenance (EIM). After the intervention, posttest mean scores increased to 48.50 for the control group and 61.30 for the experimental group, both of which were qualitatively described as average mastery.

□ All competency areas were identified as least mastered during the pretest for both groups, with overall mean scores falling within the 15.00–34.999 range categorized by the Department of Education as low mastery. Following instruction, mastery levels in all competencies improved to average, as indicated by the posttest mean scores of both groups.

□ Statistical analysis showed no significant difference between the pretest mean scores of the control and experimental groups ($t = 0.83$), as the computed value was lower than the critical value of 2.002 at the 0.05 level of significance with 58 degrees of freedom.

□ A significant difference was found between the posttest mean scores of the control and experimental groups ($t = 5.43$), which exceeded the critical value of 2.002 at the 0.05 level of significance, indicating that the use of the electrical wiring board had a positive effect on students' academic performance.

□ A significant improvement was observed between the pretest and posttest mean scores of the control group ($t = 15.81$) and the experimental group ($t = 21.53$). Both computed values were greater than the critical value of 2.04 at the 0.05 level of significance, confirming that learning gains occurred in both groups, with greater improvement in the experimental group.

Discussions

1. The findings revealed that both the control and experimental groups initially demonstrated low mastery in Electrical Installation and Maintenance; however, their posttest performance improved to an average level of mastery after instruction.
2. The least mastered competencies in the pretest for both groups were mensuration and calculation, interpretation of technical drawings and plans, occupational health and safety practices,

maintenance of tools and equipment, and preparation of electrical supplies, materials, and tools. In the posttest, mastery levels in these competencies improved to an average level for both groups.

3. There was no significant difference between the pretest mean scores of the control and experimental groups, indicating that the two groups had comparable entry-level competencies before the intervention.
4. A significant difference was observed between the pretest and posttest mean scores of the experimental group, demonstrating the effectiveness of the electrical wiring board in enhancing students' academic performance.
5. A significant difference was likewise found between the pretest and posttest mean scores of both the control and experimental groups, with the experimental group showing greater learning gains than the control group.
6. Electrical wiring boards or circuit trainers should be designed and utilized to specifically address the least mastered competencies in Electrical Installation and Maintenance.
7. EIM teachers are encouraged to integrate the use of electrical wiring boards or circuit trainers into instruction to improve students' academic performance and to help develop globally

competitive, skilled, and productive learners guided by principles of equity.

8. The electrical wiring board or circuit trainer developed in this study may be patented, reproduced, and adopted by both public and private schools offering EIM programs.
9. School administrators should provide adequate funding and institutional support for the development, acquisition, and use of circuit trainers in EIM subjects.
10. The Department of Education (DepEd) may support teachers and researchers in public and private schools by providing the necessary tools, materials, and resources for the development of circuit trainers and similar instructional innovations.
11. The Technical Education and Skills Development Authority (TESDA) may conduct seminars, in-service trainings, and competency assessments at the division level to enhance teachers' skills in developing and utilizing circuit trainers and other instructional devices.
12. TLE/T.H.E. teachers and future researchers are encouraged to conduct similar studies with larger samples and in different settings to further validate and strengthen the findings of this research.

Part II

Academic

Articles

Green Chemistry: Advancing Science Through Sustainability

Cristopher G. Reyes

Concept and Significance of Green Chemistry

Conventional chemical practices often depend on toxic raw materials, generate substantial waste, and require high energy consumption. Green chemistry seeks to address these issues by integrating environmental considerations directly into the design stage of chemical processes. Its primary goal is the prevention of pollution at the source rather than remediation after damage has occurred. This proactive strategy makes green chemistry a vital component of sustainable development.

The significance of green chemistry lies in its ability to safeguard human health, preserve ecosystems, and optimize the use of natural resources. By encouraging safer chemical alternatives and more efficient processes, green chemistry supports innovation across multiple industries without compromising environmental integrity.

The Twelve Principles of Green Chemistry

The foundation of green chemistry is built upon the twelve principles formulated by Paul Anastas and John Warner. These principles provide a comprehensive framework for developing safer and more sustainable

chemical practices. Key principles include waste prevention, the use of renewable feedstocks, the design of less toxic chemical products, energy efficiency, and the reduction of derivatives and auxiliary substances such as solvents.

The application of these principles enables chemists to improve process efficiency while reducing environmental impact. For instance, the use of biodegradable materials and environmentally benign catalysts exemplifies how green chemistry principles are translated into practical solutions.

Applications of Green Chemistry

Green chemistry has wide-ranging applications across various sectors. In agriculture, it supports the development of eco-friendly fertilizers and pesticides that reduce soil degradation and water contamination. In the pharmaceutical industry, green chemistry improves synthesis methods by minimizing hazardous waste and enhancing production efficiency. Consumer products such as biodegradable detergents, sustainable packaging materials, and plant-based cleaning agents further demonstrate its everyday relevance.

Additionally, green chemistry plays a crucial role in advancing renewable energy technologies, including biofuels and sustainable materials for energy storage and conversion, thereby reducing reliance on fossil fuels.

Benefits to Society and the Environment

The adoption of green chemistry offers significant environmental, economic, and social benefits. Environmentally, it reduces pollution and conserves natural resources. Economically, it lowers manufacturing costs through efficient material and energy use. Socially, it enhances workplace safety and public health by reducing exposure to toxic substances.

Furthermore, green chemistry contributes to global sustainability initiatives by supporting responsible consumption and production, clean energy development, and climate change mitigation.

Conclusion

Green chemistry represents a paradigm shift in chemical science by aligning innovation with sustainability. Through the integration of environmentally responsible practices in education, research, and industry, green chemistry enables society to meet present needs without compromising the ability of future generations to thrive. As global demand for sustainable solutions continues to grow, green chemistry will remain essential in shaping a safer, cleaner, and more resilient world.

Effects Of Google on Learning and Information Behavior

Cristopher G. Reyes

Google has become an integral part of daily life, particularly in the field of education, where it serves as a primary source of information for students and educators. As a widely used search engine, Google enables users to access a vast range of academic, scientific, and general knowledge resources within seconds. Its convenience and accessibility have significantly transformed learning behaviors and information-seeking practices.

One of the major positive effects of Google is its contribution to independent and flexible learning. Students are able to clarify lessons, conduct research, and explore topics beyond classroom discussions. Google also enhances digital literacy by encouraging users to evaluate multiple sources, compare perspectives, and develop research skills. These features support learner autonomy and promote continuous learning.

However, excessive dependence on Google may result in negative learning outcomes. The ease of accessing information can reduce deep cognitive processing, critical thinking, and long-term memory retention. This phenomenon, often referred to as the “Google Effect,” suggests that individuals are more likely to remember where to find information rather than the information itself. Furthermore, the presence of inaccurate or

misleading online content poses challenges, particularly for users who lack strong information evaluation skills.

In summary, Google has a significant impact on learning and information behavior. While it offers substantial educational benefits, its effectiveness depends on responsible use and the development of critical thinking and digital literacy skills.

Effects of Gossip on Social Relationships and Behavior

Gossip is a common social behavior that involves discussing information about individuals who are not present. It occurs in various social settings, including schools, workplaces, and communities, and plays a role in shaping interpersonal relationships and group dynamics. Gossip can have both constructive and harmful effects depending on its content and intention.

On the positive side, gossip can strengthen social connections and promote group cohesion. It allows individuals to share experiences, understand social norms, and build trust within a group. In some instances, gossip serves as a means of social regulation by discouraging undesirable behavior and warning others of potential risks.

Conversely, negative gossip can lead to serious social and emotional consequences. The spread of rumors or false information can harm reputations, lower self-esteem, and cause emotional distress. In educational settings, gossip may contribute to bullying, social exclusion, and a decline in academic engagement. Prolonged exposure to negative gossip can also create a hostile environment that affects overall mental well-being and social harmony.

In conclusion, gossip is a powerful form of social communication with both positive and negative implications. Encouraging ethical communication, empathy, and respect is essential in reducing the harmful effects of gossip while fostering healthy social relationships.

Teachers and Parents: A Dynamic Partnership

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Abstract

A partnership between home and school is indeed formidable is characterized by genuine sharing and collaboration between teachers and parents. The parents have their children's best interests at heart and there seems no limit to what they are willing to harness within their resources. The teachers equally feel attached and concerned with the children and are ready to offer their utmost effort and time, even beyond the confines of the classroom. Every child under their care is center and there is nothing so time consuming, so attention-demanding and physically-draining that they will not do. The benefits derived from a free flow of useful and relevant information between the home and the school cannot be over emphasized. One cannot fully achieve its goals without the help of the other. Nothing is impossible if both are sincerely willing to share responsibilities and commit their time, effort and resources for such a noble investment-child growth and nurturance.

It is recognition of the tremendous benefits both parties stand to gain that an attempt to situate theme in real, easy to-attain alliance is hereby offered.

Intimate observations of parents in the home can serve as inputs in planning learning tasks. Easily, the mother can communicate to the teacher certain behaviors that need to be corrected. Lack of initiative in doing homework regularly can be traced to unchallenging homework or inconsistency in checking their assignments everyday. The information they sometimes are required to gain through home reading may not be accomplished if there are not many children's books at home. Gathering of data through exploration in the immediate environment might be a more effective methodology. A system of rewards will definitely generate eagerness in tackling home investigations.

Respect for elders is a commendable trait that could be modelled and internalized at a young age in the home. Carried to the school they will continue to manifest consideration and high regard for the right of others. The teachers in return could exhibit the same respect for everyone.

Some unusual interest of the child that are observed at home are worth communicating to those who plan the daily activities. The inclination to music or poetry can be part of the motivational strategies. Special talents such as sketching and painting can be nurtured by both the home and the school through activities that expose them to nature. Inventiveness and

originality are values that can be developed during their formative years and much depend on both parents and teachers' sensitivity to early manifestations.

Learning to cooperate starts from the home and is enhanced in a conducive school atmosphere. Today, one of the organizational techniques being employed, whether in class discussions or group experimentations, is one termed cooperative learning. In a group of 5 or 6, a member is responsible for a part of the lesson and later they meet to discuss or perform the experiment as a group. Such cooperative attitude can be reinforced in similar activities wherein members of the family share responsibilities for a particular task.

How to Manage and Lead by Examples

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The key to effective leadership lies on how you practice what you preach. Leadership by Example is the true expression of ourselves – our thoughts, values, attitudes, feelings and beliefs. It is the expression of our positive feeling, values and attitudes of others. Leading by example is learning to ask yourself: How can I make it clear that this is important to me? The important word is “I”. You cannot give what you do not have. You cannot manage others. If you cannot manage yourself. You cannot discipline other, unless you discipline yourself. You cannot control others unless you control yourself. Control is a very important part of Leadership by Example. It is important from the beginning to learn that you do have control. Action speaks louder than words.

Leadership by example is based on the premises that goodness is diffusive by itself. “Bonum diffusivum sui”. Man has the right to be good and to spread that goodness he has in himself. Leadership By Example recognizes

and builds on strengths so, that goodness may grow, shine and may be established. A leader by example continuously professionalizes and self-develop so that he may grow in goodness and righteousness. His good examples continuously challenge his people so that goodness may shine. He reinforces the positive behavior may stay.

A supervisor must possess the essential managerial skills which are as follows: 1) General management skills, 2) Manufacturing/production management skills and 3) Specialized knowledge and skills in manufacturing.

The general management skills consist of the following:

1. Planning
 - a. For improvement
 - b. To meet specific new requirements
 - c. To anticipate changing conditions and be geared to overall organizations goals.
2. Organizing and integrating
 - a. A program
 - b. Day-to-day operations
3. Executing
 - a. A program
 - b. Day-to-day operations.

Reference: Andres, Tomas Quintin Donato, 1993.

Leadership

By Example, Manila

Our Lady of Manaoag Publisher

Values Formation

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Region 1

Among the values and attitudes greatly expected of a teacher are:

1. Love and care for the young

For the teacher the young symbolize innocence, purity and goodness. Such traits are nurtured by one who is equally kindhearted and young in spirits. As the old sayings goes, “Love of the same magnitude” among the lovable ones.

2. Honesty and Sincerity

An honest and sincere teacher serves as the best model in developing the same among the young. To uphold the truth is the apex of one’s character worth emulating. No doubt, a sincere and unpretentious teacher could easily win the young. Such a teacher behavior is contagious.

3. Open-mindedness and Critical-Mindedness

An open-minded teacher respects the rights of others, considers other people's feelings and looks at all sides of an issue. She is not dogmatic, is willing to listen to others and can accept criticisms and suggestions.

A critical minded teacher analyzes sources of information, and weighs evidences. With children association daily with these ideal teachers, their objective and scientific upbringings is assured.

4. Curiosity and an Inquiring Mind

A curious teacher can easily make children equally curious. Curiosity that is aroused and sustained can lead the children to ask questions about things and happenings in their environment. Bright and inquisitive teacher makes children grow with an eagerness for knowledge and answers their endless questions. Consistent and intelligent inquiries are the ingredients for life long pursuits, be it academic or otherwise.

5. Creativity and Flexibility

A creative teacher thinks of new and original ways of doing things. She keeps on trying better and better ways of teacher instead of the boring routine. In so doing, children likewise become flexible and

they attempt other modes in finding answer and information rather than follow the traditional procedures. In the long run, they are able to generate now and sometimes divergent ideas, device new tools and improve on ways of solving problems. A highly creative teacher is an asset and serves as permanent assurance in developing creative children.

6. Persistence

He continues to serve beyond the limit of time, space and resources. Unmindful of rewards, he pursues to completion assigned tasks. Children observing a persistent teacher who remains interested what he is doing despite constraints and personal difficulties will surely be moved to behave in the same manner. A beautiful saying for persistent teachers goes “They walk an extra mile.”

7. Responsibility

A highly responsible teacher keeps his words and take upon himself the completion of their own tasks. They are fully aware of his duties and obligations to be observed accomplished, willing to accept and own failures and other consequences of own actions.

A strong sense of responsibility, coupled with honesty is an adulthood. It is therefore important that they grow amidst responsible teachers.

8. Humility

A humble teacher recognized and accepts own weaknesses and limitations. He is then ready to listen to criticisms, willing to consult others and can change own ideas in the light of group decision. Great educators, inventors and remarkable leaders are humble. Teachers can learn from their autobiographies, in turn make them as their exemplars. Then, their teacher image will not be forgotten by their students for a long time Bragging about one's achievements has never gone into their heads.

Reference:

Salandanan, Gloria G. 2001. Teacher
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On Sustainable Development Goal Issue

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The Sustainable Development Goals are series of measures to end poverty, protect the planet and ensure the global well-being of human beings. Do you know what they are and how they were conceived? Human-kind faces many challenges to ensure that all people have the same development and well-being opportunities.

In a global perspective, in the year 2000, the United Nations adopted eight (8) objectives to meet the main needs of the poorest, the Millennium Development Goals: Eradicate extreme poverty and hunger. Achieve universal primary education; Promote gender equality; Reduce child mortality; Improve maternal health; Combat HIV/AIDS, Malaria and other diseases; Ensure environmental sustainability; and develop a global partnership for development.

These are ambitious goals, disaggregated into targets, that require the collaboration of civil society and the public and private sectors, whose

success means a more egalitarian and livable world. But the question to be answered is- What do the Sustainable Development Goals seek?

The Sustainable Development Objectives cover different aspects of social development, environmental protection and economic growth, and these are the main ones: The eradication of poverty and hunger so as to ensure a healthy life; Achieve universal access to basic services such as water, sanitation and sustainable energy; Support the generation of development opportunities through inclusive education and decent work; Foster innovation and resilient infrastructures by creating communities and cities capable of producing and consuming in a sustainable way; Reduce inequalities in the world, especially gender inequalities; Caring for the environment by combatting climate change and protecting oceans and terrestrial ecosystems; Promote collaboration between different social actors to create an environment of peace and sustainable development.

If we achieve this sustainable development goals among people, government and companies, we will make the world a more prosperous place for all. In the last few years, it's clear that issues like climate change, plastic waste management, increasing migration, and other major issues are growing global concerns. For the last two years students have become

highly active in protesting and finding solutions to both national and global issues.

As a teacher, traditional education gives students few opportunities to explore, understand, and solve real world problems. So how are students expected to prepare for all these global issues? The SDGs can be used as a framework and tool to support students develop their research, critical thinking, problem solving, teamwork, and communication skills. Students need this 21st Century Skills to prepare them for the major real-world problems they will face during their lifetime. Teaching Sustainable Development Goals and exploring them can also support students find their identity and purpose.

Introducing Gender Equality in the Lower Grades

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The second part addresses the issue of declining academic performance among students, arguing that no single individual or group should be blamed. Instead, it highlights the collective responsibility of teachers, schools, parents, and learners. Effective teaching strategies, adequate learning resources, parental support, and student accountability are all essential factors in improving educational outcomes. The article concludes that through collaboration, commitment, and shared responsibility, both gender equality awareness and improved academic performance can be achieved, leading to holistic student development.

All humans are born equal no matter their gender. It is our duty to make sure that women and men have equal opportunities and rights in every part of life.

Gender in general is the range of characteristics pertaining to, and differentiating between masculinity and femininity. Depending on the context, these characteristics may include biological sex (i.e. the state of

being male, female or an intersex variation), sex-based social structures (i.e. gender roles), or gender identity. Most cultures use a gender binary, having two genders (boys/men and girls/women); those who exist outside these groups fall under the umbrella term non-binary or genderqueer. Some societies have specific genders besides “man” and “woman” , such as the hijras of South Asia; these are often referred to as third genders (and fourth genders, etc.) Gender equality is achieved when men and women enjoy the same rights and opportunities across all sectors of society, including economic participation and decision-making, and when the different behaviors, aspirations and needs of women and men are equally valued and favored. Gender equality is not only a fundamental human right but a necessary foundation of a peaceful, prosperous and sustainable world.

GENDER EQUALITY maybe a high-sounding term for children of the kindergarten to third grade. But the quality of life in the future depends upon beginning early to understand how each person is an integral part of their community. Teachers are aware of this importance. The teacher should introduce this concept at the level of the child’s understanding, maybe in a small segment, one at a time.

Since gender equality has assumed growing importance lately, our children in the primary grades should be involved in activities on this subject,

meaningful to them and suitable as a foundation for their further understanding.

In the lower grades, the activities to be planned should be preferably be first-hand experiences on such topics like 1) Proper Health/Grooming, 2) School Rules & Regulations including Directional signages posted in conspicuous places like “This way to the “HE/MALE, SHE/FEMALE comfort room, 3) Gender Awareness/Equality to be integrated in their lessons, 4) Boys’ and Girls’ Rights/Children’s Rights, and 5) Child Protection Policy/Anti-bullying Act

It is said that the school is the first place to demonstrate to young children how they may avail themselves of gender-friendly and respectful environment. Has the school a source of full protection against bullying? Young children need many opportunities to integrate an consolidate school activities with those in the home. Most of the topics mentioned are learned in school but it is practically noted that the first learnings in school but it is practically noted that the first learnings were already taught at home.

Poor Academic Performance, Who's To Be Blamed?

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It is a common observation among DepEd workers, especially the classroom Teachers, the decline of pupil's academic performance as reflected by poor results of written and performance tasks. The big questions are: Who is to be blamed? Who is at fault? Who has the responsibility of addressing the issue?

As we going to point our finger to the teacher as the key factor in classroom learning? We normally ask the questions: Does he give remedial teaching to improve academic performance of pupils who are slow? How good is he in managing the class? Does he use varied teaching styles that cater to the individual needs of the pupils? Does he show a genuine love to his pupils?

If not the teacher, are we going to reprehend the school as provider of formal education? We often ask: Are our schools provided with various reading materials that pupils can use, especially textbook which is K to 12

compliant? Does it provide proportionate learning material to pupils? Does it offer conducive environment for learning? Is it really Kto12 ready?

Or probably can we condemn the parents and say: Do they monitor their child's performance from their teachers? Do they provide the financial and moral support to them? Do they discuss with them what they have learned in school? Do they follow up what the child has learned in the school?

Or is the failure caused by the pupil himself? Considered as the center of the educative process, does he participate and interact when the teacher gives learning activities? Or he just totally engrossed with online games that robbed his time in reviewing his lesson? Does he display the right attitude to achieve effective teaching-learning situation?

Stop pointing your finger and passing the blame to each other because no single person is at fault. Instead, each should do his own responsibility. Bear in mind that the pupils' performance is an effort of the child himself and the people around him.

To our teachers, continue pouring your love to your pupils. When we love them, we tend to do everything just for them to become a better person. Despite the lapses of our department, like lack of textbooks and other

learning resources, we can still make a difference into the lives of our learners.

On the part of our parents, your support is very much needed nowadays. Joining PTA meetings and coming to school to monitor your child's performance are simple ways to show your support and concern to him. Teach them, too, good values so they can acquire more wisdom and knowledge in school.

To our pupils, be more responsible in your studies. Whatever good plans you have in the future, start working for it now. When you succeed, that is for your own benefit. But if you fail, you will suffer the consequence.

Again, no one is to be blamed, no one is at fault, with the different processes that the teacher, school, parent and pupils undergoes, a beneficial effect is expected. Effective teaching, challenging environment, supportive parent and responsible pupil will yield a better pupil's academic performance. This may not happen immediately, but through the joint effort and commitment of everyone, eventually the desired outcome will be achieved.

Risk and Responsibility in Leading by Example

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Ronald Lippit, The “Journal of Applied Behavioral Science, Vol. 18, No. 3 pp, 395-403). Enumerate six trends that affect the changes in leader and follower roles. These results into the changing leader-followership. The six trends are the following:

1. Double bind of maintaining quality with reduce resources-The leaders improve the quality of reduces resources and the followers are responsible and loyal in the job and reduced resources.
2. Expectations of shared power and responsibility-Leaders expect initiative from the followers. In some cases, duties are separated and sometimes put together but these become a problem to both groups. Leaders become doubtful and fear loss of control and followers don’t want the new task because they lack skill and support;
3. Interdependence, collaboration and communication-one of the results of decentralized organizational functioning is the interdependence and collaboration between the manager and the supervisor.

4. More people required for problem solving-In solving problems more peoples are needed to tackle the task, particularly the right people for the right time. It requires flexible or strong leaders as followers with great patience.
5. Integrating technological and human resources-It is mixing together of the use of computers as technological resources and people as human resource.
6. Reorientation, renewal and new competency development-Changeability is need because its feelings, continue the role of the learner or have a new task for oneself, invest and support of training programs for followers.
 - A. Attitudes and skills of proactive response to changes-One attitude to these changes is to defend equitable, research and develop and sacrifice personnel.
 - B. Mobilizing resources for the self-This requires developing special skills in listening to internal dialogues between positive and negative feelings, support for caution and reflecting alternatives.
 - C. Achieving a balance of task work focus and process work focus-If groups are engaged in production and problem solving with each other's.

- D. Divers – person team building-Leaders have to use some procedure for identifying, recording and striving the skills, competence and training of persons they lead.
- E. Teaming people and technology-the two must understand and appreciate each other so the leader has to involve procedures for teaching both of them.
- F. Leaders have neighbors and outsiders as potential resources- A leader must share and exchange resources in order to survive the environment. Securing resources need documentation and evaluation.

Reference:

Andres, Tomas Quintin Donato. 1993. Leadership By Example, Manila: Our Lady of Manaoag Publishers.

Transformational Leaders in the New Educational Scheme

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Leadership is hard work, and it's getting harder. To truly excel, to be truly a great leader over the long term, you must have the courage and persistence to do the hard work of leadership. Delivering consistent financial results, attaining high team performance, executing strategy, managing multiple and often conflicting priorities, and driving innovation isn't easy. You must realize that there is hard work that you alone as a leader can and must do, and if you don't do it, you'll set yourself and your organization back.

However, in my experience I find many leaders who are reluctant to dig in and choose instead to avoid the hard work. Others act as bystanders, watching things happen. And then there are those leaders who simply look for the easy way out, thinking that some quick fix idea will make everything better, only to be surprised when these quick fixes don't fix anything. We need to stop underestimating what it takes to be a consistently great leader.

Sure, leadership can be easy if you're satisfied with mediocrity. But that's not what the leadership contract is all about.

It's time that we as leaders understand that the real work of leadership isn't easy. We also need to understand that we'll need resilience, determination, and a deep sense of personal resolve to be effective. This is what the third term of the leadership contract is all about. Evolving how you lead-you may already sense-is hard work. It's hard for multiple reasons. It taxes an ability that most of us have in limited supply, which is being able to see ourselves objectively. People who have studied and worked with thousands of leaders-people like Robert Kegan and Keith Eigel-help us understand that developing this ability to put ourselves in perspective is how we progress as leaders.

Transforming how you lead is hard because leadership is a curious amalgam, different in nearly everyone, and made up of ideas, skills, and behaviors we may not know the origin or validity of. Finally, transforming how you lead is difficult because leadership has become, more than ever, a team sport. A leadership team's ability to become more adaptive requires not just individual change, but collective and coordinated change. Great transformation will be observable.

School Principal's Impact Towards Learners' Scholastic Achievement

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Many researchers have confirmed that school principals have a significant impact on learners' achievement. Without question, an effective principal is the key to a successful school.

School principals are responsible for the overall functioning of their school. As managers, they oversee the day-to-day operations of schools. They set budgets, manage the school facility, and develop relationships with the broader community. As instructional leaders, they direct and supervised the development, delivery, assessment, and improvement of educational programs and activities in their schools.

They also hire or make recommendations to district administrators and induct new teachers, provide professional development opportunities to teachers, evaluate their performance, and assign to classrooms.

Additionally, they act as the liaison between the district and the school itself, interpreting and implementing policies passed down by federal, state or district authorities, then communicating feedback on the success of those initiatives (Dhuey & Smith, 2018).

In sum, school principal's practices influence school conditions, teacher quality, instructional quality, and learner's achievement. Research also indicates that effective school principals have significant effects on learner absenteeism, learner engagement with learner's academic self-efficacy, staff satisfaction, and collective teacher efficacy (e.g., Dou, Devos & Lalcke, 2016; Hitt & Tucker, 2015; Loeb, Kalogrides & Beteille, 2012).

School Principal's Direct Impact on Learner Achievement Research has found that the impact of highly effective school principal is equivalent to raising the achievement of a typical learner by two or more months of extra learning in a single school year, whereas ineffective school principals' lower achievement by the same amount (Branch, Hanushek & Rivkin, 2012, 2013).

There is substantial variation when it comes to effectiveness across school principal's response towards academic needs of the learners. Furthermore,

the school principal contributed much in terms of learning and performance of the learners in an area of scholastic achievement.

School Principal's Impact Towards Learners' Scholastic Achievement

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Schools need leaders who have a vision for improving quality and learning outcomes, and who are also effective at ongoing management tasks.

School managers can positively contribute to school effectiveness when they are prepared and able to use extensive leadership knowledge to solve complex school-based problems, and to build trust through working relationships with school staff, parents, learners, and the community.

The success of any depends much on the ability of its leader and the support of its staff. Leadership, in all its styles and types, if it were to succeed must be adhered to things that are to work on with and not the will to work, and that the human environment is an important part of what the members and their leader work with, a need to keep up the work situation sustainable and progressive. In the task of teaching, the role of leadership plays a significant mark.

To be a leader, you have to show profound qualities in yourself and be able to lead yourself at first. If you are not able to lead yourself it would be difficult for you to inspire others all the way. You should be fully capable

of leading yourself and thus inspiring others towards that particular position, which is the goal. The achieving of inner balance always requires being clear about yourself fully and the targets you wish to achieve.

School leaders should aim high to produce top rank learners that could compete in any other schools in the country.

A leader with vision has a clear, vivid picture of where to go, as well as a firm grasp on what success looks like and how to achieve it. But it's not enough to have a vision, leaders must also share it and act upon it. People sometimes are unable to manage the things due to lack of leadership skills. The major decisions of life are not been settled positively due to the deficiency of mindset in this particular area. The block in personality needs special assistance to overcome these flaws which hinder to progress in career or life. These blocks can be eliminated by proper assistance to overcome the fears and doubts developed over the years.

Modern day education does not only look at the teachers as the key imparter of knowledge but prescribes upon him a multi-pronged function such as a guidance counselor, a second parent to the children under his care, a model expressive of perfection, excellence and dignity, a problem identifier and problem solver and to emphasize, as a leader.